An n-connected map induces an equivalence on n-truncations

The Univalent Foundations Program

hott-book:578b85cc8d586b1677ec4335148adeb443057d24; CC BY-SA 3.0

Editorial note. Original author statement, proof and supporting context preserved. Only the document wrapper and navigational reference bindings are added; no mathematical repair or correctness certification.

Allowed standard prerequisites (editorial).

- Homotopy type theory. Univalence, dependent products and sums, identity types, propositional and n-truncation; native book references retained.. Standard background, without replacing the counted argument.

Source reference. Original reference 7.6: combinatorics-logic/sources/hott-book/main.labelnumbers.first-edition, source lines 1349–1349. Reference to book navigation or standard prerequisite; the target author proof is retained in full.

Source reference. Original reference 4.4: combinatorics-logic/sources/hott-book/main.labelnumbers.first-edition, source lines 1414–1414. Reference to book navigation or standard prerequisite; the target author proof is retained in full.

Source reference. Original reference 4.7.6: combinatorics-logic/sources/hott-book/main.labelnumbers.first-edition, source lines 1419–1419. Reference to book navigation or standard prerequisite; the target author proof is retained in full.

Source reference. Original reference 4.7.3: combinatorics-logic/sources/hott-book/main.labelnumbers.first-edition, source lines 1532–1532. Reference to book navigation or standard prerequisite; the target author proof is retained in full.

Source reference. Original reference 8.6: combinatorics-logic/sources/hott-book/main.labelnumbers.first-edition, source lines 1656–1656. Reference to book navigation or standard prerequisite; the target author proof is retained in full.

Source reference. Original reference 4.6: combinatorics-logic/sources/hott-book/main.labelnumbers.first-edition, source lines 1680–1680. Reference to book navigation or standard prerequisite; the target author proof is retained in full.

Source reference. Original reference 7.3.9: combinatorics-logic/sources/hott-book/main.labelnumbers.first-edition, source lines 1906–1906. Reference to book navigation or standard prerequisite; the target author proof is retained in full.

Source reference. Original reference 7.3.2: combinatorics-logic/sources/hott-book/main.labelnumbers.first-edition, source lines 1907–1907. Reference to book navigation or standard prerequisite; the target author proof is retained in full.

7.1 Connectedness

An n-type is one that has no interesting information above dimension n. By contrast, an *n-connected type* is one that has no interesting information *below* dimension n. It turns out to be natural to study a more general notion for functions as well.

Definition 7.1.1. A function $f : A \to B$ is said to be **n-connected** if for all $b : B$, the type $\|\mathsf{fib}_f(b)\|_n$ is contractible:

$$\mathsf{conn}_n(f) :\equiv \prod_{b:B} \mathsf{isContr}(\|\mathsf{fib}_f(b)\|_n).$$

A type A is said to be **n-connected** if the unique function $A \to \mathbf{1}$ is n-connected, i.e. if $\|A\|_n$ is contractible.

Thus, a function $f : A \to B$ is n-connected if and only if $\mathsf{fib}_f(b)$ is n-connected for every $b : B$. Of course, every function is (-2)-connected. At the next level, we have:

Lemma 7.1.2. *A function f is (-1)-connected if and only if it is surjective in the sense of original reference 4.6.*

Proof. We defined f to be surjective if $\|\mathsf{fib}_f(b)\|_{-1}$ is inhabited for all b. But since it is a mere proposition, inhabitation is equivalent to contractibility. □

Thus, n-connectedness of a function for $n \geq 0$ can be thought of as a strong form of surjectivity. Category-theoretically, (-1)-connectedness corresponds to essential surjectivity on objects, while n-connectedness corresponds to essential surjectivity on k-morphisms for $k \leq n + 1$.

Lemma 7.1.2 also implies that a type A is (-1)-connected if and only if it is merely inhabited. When a type is 0-connected we may simply say that it is **connected**, and when it is 1-connected we say it is **simply connected**.

Remark 7.1.3. While our notion of n-connectedness for types agrees with the standard notion in homotopy theory, our notion of n-connectedness for *functions* is off by one from a common indexing in classical homotopy theory. Whereas we say a function f is n-connected if all its fibers are n-connected, some classical homotopy theorists would call such a function $(n + 1)$-connected. (This is due to a historical focus on *cofibers* rather than fibers.)

We now observe a few closure properties of connected maps.

Lemma 7.1.4. *Suppose that g is a retract of a n-connected function f. Then g is n-connected.*

Proof. This is a direct consequence of original reference 4.7.3. □

Corollary 7.1.5. *If g is homotopic to a n-connected function f, then g is n-connected.*

Lemma 7.1.6. *Suppose that $f : A \to B$ is n-connected. Then $g : B \to C$ is n-connected if and only if $g \circ f$ is n-connected.*

Proof. For any $c : C$, we have

$$\begin{aligned}
\|\mathsf{fib}_{g\circ f}(c)\|_n &\simeq \Big\| \sum_{w:\mathsf{fib}_g(c)} \mathsf{fib}_f(\mathsf{pr}_1 w) \Big\|_n && \text{(by original reference 4.4)} \\
&\simeq \Big\| \sum_{w:\mathsf{fib}_g(c)} \|\mathsf{fib}_f(\mathsf{pr}_1 w)\|_n \Big\|_n && \text{(by original reference 7.3.9)} \\
&\simeq \|\mathsf{fib}_g(c)\|_n. && \text{(since } \|\mathsf{fib}_f(\mathsf{pr}_1 w)\|_n \text{ is contractible)}
\end{aligned}$$

It follows that $\|\mathsf{fib}_g(c)\|_n$ is contractible if and only if $\|\mathsf{fib}_{g\circ f}(c)\|_n$ is contractible. □

Importantly, n-connected functions can be equivalently characterized as those which satisfy an "induction principle" with respect to n-types. This idea will lead directly into our proof of the Freudenthal suspension theorem in original reference 8.6.

Lemma 7.1.7. *For $f : A \to B$ and $P : B \to \mathcal{U}$, consider the following function:*

$$\lambda s.\, s \circ f : \Big(\prod_{b:B} P(b)\Big) \to \Big(\prod_{a:A} P(f(a))\Big).$$

For a fixed f and $n \geq -2$, the following are equivalent.

(i) f is n-connected.
(ii) For every $P : B \to n\text{-}\mathsf{Type}$, the map $\lambda s.\, s \circ f$ is an equivalence.
(iii) For every $P : B \to n\text{-}\mathsf{Type}$, the map $\lambda s.\, s \circ f$ has a section.

Proof. Suppose that f is n-connected and let $P : B \to n\text{-}\mathsf{Type}$. Then we have the equivalences

$$\begin{aligned}
\prod_{b:B} P(b) &\simeq \prod_{b:B} \Big(\|\mathsf{fib}_f(b)\|_n \to P(b)\Big) && \text{(since } \|\mathsf{fib}_f(b)\|_n \text{ is contractible)} \\
&\simeq \prod_{b:B} \Big(\mathsf{fib}_f(b) \to P(b)\Big) && \text{(since } P(b) \text{ is an } n\text{-type)} \\
&\simeq \prod_{(b:B)} \prod_{(a:A)} \prod_{(p:f(a)=b)} P(b) \\
& && \text{(by the left universal property of } \Sigma\text{-types)} \\
&\simeq \prod_{a:A} P(f(a)). && \text{(by the left universal property of path types)}
\end{aligned}$$

We omit the proof that this equivalence is indeed given by $\lambda s.\, s \circ f$. Thus, (i)⇒(ii), and clearly (ii)⇒(iii). To show (iii)⇒(i), consider the type family

$$P(b) :\equiv \|\mathsf{fib}_f(b)\|_n.$$

Then (iii) yields a map $c : \prod_{(b:B)} \|\mathsf{fib}_f(b)\|_n$ with $c(f(a)) = \big|(a, \mathsf{refl}_{f(a)})\big|_n$. To show that each $\|\mathsf{fib}_f(b)\|_n$ is contractible, we will find a function of type

$$\prod_{(b:B)} \prod_{(w:\|\mathsf{fib}_f(b)\|_n)} w = c(b).$$

By original reference 7.3.2, for this it suffices to find a function of type

$$\prod_{(b:B)} \prod_{(a:A)} \prod_{(p:f(a)=b)} |(a,p)|_n = c(b).$$

But by rearranging variables and path induction, this is equivalent to the type

$$\prod_{a:A} \left|(a, \mathsf{refl}_{f(a)})\right|_n = c(f(a)).$$

This property holds by our choice of $c(f(a))$. □

Corollary 7.1.8. *For any A, the canonical function $|-|_n : A \to \|A\|_n$ is n-connected.*

Proof. By original reference 7.3.2 and the associated uniqueness principle, the condition of Lemma 7.1.7 holds. □

For instance, when $n = -1$, Corollary 7.1.8 says that the map $A \to \|A\|$ from a type to its propositional truncation is surjective.

Corollary 7.1.9. *A type A is n-connected if and only if the map*

$$\lambda b.\, \lambda a.\, b : B \to (A \to B)$$

is an equivalence for every n-type B. In other words, "every map from A to an n-type is constant".

Proof. By Lemma 7.1.7 applied to a function with codomain $\mathbf{1}$. □

Lemma 7.1.10. *Let B be an n-type and let $f : A \to B$ be a function. Then the induced function $g : \|A\|_n \to B$ is an equivalence if and only if f is n-connected.*

Proof. By Corollary 7.1.8, $|-|_n$ is n-connected. Thus, since $f = g \circ |-|_n$, by Lemma 7.1.6 f is n-connected if and only if g is n-connected. But since g is a function between n-types, its fibers are also n-types. Thus, g is n-connected if and only if it is an equivalence. □

We can also characterize connected pointed types in terms of connectivity of the inclusion of their basepoint.

Lemma 7.1.11. *Let A be a type and $a_0 : \mathbf{1} \to A$ a basepoint, with $n \geq -1$. Then A is n-connected if and only if the map a_0 is $(n-1)$-connected.*

Proof. First suppose $a_0 : \mathbf{1} \to A$ is $(n-1)$-connected and let B be an n-type; we will use Corollary 7.1.9. The map $\lambda b.\, \lambda a.\, b : B \to (A \to B)$ has a retraction given by $f \mapsto f(a_0)$, so it suffices to show it also has a section, i.e. that for any $f : A \to B$ there is $b : B$ such that $f = \lambda a.\, b$. We choose $b :\equiv f(a_0)$. Define $P : A \to \mathcal{U}$ by $P(a) :\equiv (f(a) = f(a_0))$. Then P is a family of $(n-1)$-types and we have $P(a_0)$; hence we have $\prod_{(a:A)} P(a)$ since $a_0 : \mathbf{1} \to A$ is $(n-1)$-connected. Thus, $f = \lambda a.\, f(a_0)$ as desired.

Now suppose A is n-connected, and let $P : A \to (n-1)\text{-}\mathsf{Type}$ and $u : P(a_0)$ be given. By Lemma 7.1.7, it will suffice to construct $f : \prod_{(a:A)} P(a)$ such that $f(a_0) = u$. Now $(n-1)\text{-}\mathsf{Type}$ is an n-type and A is n-connected, so by Corollary 7.1.9, there is an n-type B such that $P = \lambda a.\, B$. Hence, we have a family of equivalences $g : \prod_{(a:A)} (P(a) \simeq B)$. Define $f(a) :\equiv {g_a}^{-1}(g_{a_0}(u))$; then $f : \prod_{(a:A)} P(a)$ and $f(a_0) = u$ as desired. □

In particular, a pointed type (A, a_0) is 0-connected if and only if $a_0 : \mathbf{1} \to A$ is surjective, which is to say $\prod_{(x:A)} \|x = a_0\|$. For a similar result in the not-necessarily-pointed case, see original reference 7.6.

A useful variation on Lemma 7.1.6 is:

Lemma 7.1.12. *Let $f : A \to B$ be a function and $P : A \to \mathcal{U}$ and $Q : B \to \mathcal{U}$ be type families. Suppose that $g : \prod_{(a:A)} P(a) \to Q(f(a))$ is a fiberwise n-connected family of functions, i.e. each function $g_a : P(a) \to Q(f(a))$ is n-connected. If f is also n-connected, then so is the function*

$$\begin{aligned} \varphi &: \Big(\sum_{a:A} P(a)\Big) \to \Big(\sum_{b:B} Q(b)\Big) \\ \varphi(a, u) &:\equiv (f(a), g_a(u)). \end{aligned}$$

Conversely, if φ and each g_a are n-connected, and moreover Q is fiberwise merely inhabited (i.e. we have $\|Q(b)\|$ for all $b : B$), then f is n-connected.

Proof. For any $b : B$ and $v : Q(b)$ we have

$$\begin{aligned} \|\mathsf{fib}_\varphi((b, v))\|_n &\simeq \Big\| \sum_{(a:A)} \sum_{(u:P(a))} \sum_{(p:f(a)=b)} p_*(g_a(u)) = v \Big\|_n \\ &\simeq \Big\| \sum_{(w:\mathsf{fib}_f(b))} \sum_{(u:P(\mathsf{pr}_1(w)))} g_{\mathsf{pr}_1 w}(u) = \mathsf{pr}_2(w)^{-1}{}_*(v) \Big\|_n \\ &\simeq \Big\| \sum_{w:\mathsf{fib}_f(b)} \mathsf{fib}_{g(\mathsf{pr}_1 w)}(\mathsf{pr}_2(w)^{-1}{}_*(v)) \Big\|_n \\ &\simeq \Big\| \sum_{w:\mathsf{fib}_f(b)} \Big\| \mathsf{fib}_{g(\mathsf{pr}_1 w)}(\mathsf{pr}_2(w)^{-1}{}_*(v)) \Big\|_n \Big\|_n \\ &\simeq \|\mathsf{fib}_f(b)\|_n \end{aligned}$$

where the transportations along $f(p)$ and $f(p)^{-1}$ are with respect to Q. Therefore, if either is contractible, so is the other.

In particular, if f is n-connected, then $\|\mathsf{fib}_f(b)\|_n$ is contractible for all $b : B$, and hence so is $\|\mathsf{fib}_\varphi((b, v))\|_n$ for all $(b, v) : \sum_{(b:B)} Q(b)$. On the other hand, if φ is n-connected, then $\|\mathsf{fib}_\varphi((b, v))\|_n$ is contractible for all (b, v), hence so is $\|\mathsf{fib}_f(b)\|_n$ for any $b : B$ such that there exists some $v : Q(b)$. Finally, since contractibility is a mere proposition, it suffices to merely have such a v. □

The converse direction of Lemma 7.1.12 can fail if Q is not fiberwise merely inhabited. For example, if P and Q are both constant at $\mathbf{0}$, then φ and each g_a are equivalences, but f could be arbitrary.

In the other direction, we have

Lemma 7.1.13. *Let $P, Q : A \to \mathcal{U}$ be type families and consider a fiberwise transformation*

$$f : \prod_{a:A} \Big(P(a) \to Q(a)\Big)$$

from P to Q. Then the induced map $\mathsf{total}(f) : \sum_{(a:A)} P(a) \to \sum_{(a:A)} Q(a)$ is n-connected if and only if each $f(a)$ is n-connected.

Of course, the "only if" direction is also a special case of Lemma 7.1.12.

Proof. By original reference 4.7.6, we have $\mathsf{fib}_{\mathsf{total}(f)}((x,v)) \simeq \mathsf{fib}_{f(x)}(v)$ for each $x : A$ and $v : Q(x)$. Hence $\left\|\mathsf{fib}_{\mathsf{total}(f)}((x,v))\right\|_n$ is contractible if and only if $\left\|\mathsf{fib}_{f(x)}(v)\right\|_n$ is contractible. □

Another useful fact about connected maps is that they induce an equivalence on n-truncations:

Lemma 7.1.14. *If $f : A \to B$ is n-connected, then it induces an equivalence $\|A\|_n \simeq \|B\|_n$.*

Proof. Let c be the proof that f is n-connected. From left to right, we use the map $\|f\|_n : \|A\|_n \to \|B\|_n$. To define the map from right to left, by the universal property of truncations, it suffices to give a map $\mathsf{back} : B \to \|A\|_n$. We can define this map as follows:

$$\mathsf{back}(y) :\equiv \|\mathsf{pr}_1\|_n(\mathsf{pr}_1(c(y))).$$

By definition, $c(y)$ has type $\mathsf{isContr}(\|\mathsf{fib}_f(y)\|_n)$, so its first component has type $\|\mathsf{fib}_f(y)\|_n$, and we can obtain an element of $\|A\|_n$ from this by projection.

Next, we show that the composites are the identity. In both directions, because the goal is a path in an n-truncated type, it suffices to cover the case of the constructor $|-|_n$.

In one direction, we must show that for all $x : A$,

$$\|\mathsf{pr}_1\|_n(\mathsf{pr}_1(c(f(x)))) = |x|_n.$$

But $\left|(x, \mathsf{refl}_{f(x)})\right|_n : \|\mathsf{fib}_f(f(x))\|_n$, and $c(f(x))$ says that this type is contractible, so

$$\mathsf{pr}_1(c(f(x))) = |(x, \mathsf{refl})|_n.$$

Applying $\|\mathsf{pr}_1\|_n$ to both sides of this equation gives the result.

In the other direction, we must show that for all $y : B$,

$$\|f\|_n(\|\mathsf{pr}_1\|_n(\mathsf{pr}_1(c(y)))) = |y|_n.$$

$\mathsf{pr}_1(c(y))$ has type $\|\mathsf{fib}_f(y)\|_n$, and the path we want is essentially the second component of the $\mathsf{fib}_f(y)$, but we need to make sure the truncations work out.

In general, suppose we are given $p : \left\|\sum_{(x:A)} B(x)\right\|_n$ and wish to prove $P(\|\mathsf{pr}_1\|_n(p))$. By truncation induction, it suffices to prove $P(|a|_n)$ for all $a : A$ and $b : B(a)$. Applying this principle in this case, it suffices to prove

$$\|f\|_n(|a|_n) = |y|_n$$

given $a : A$ and $b : f(a) = y$. But the left-hand side equals $|f(a)|_n$, so applying $|-|_n$ to both sides of b gives the result. □